



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

INNOVATIVE TEACHING PLAN

Year/Semester/Section: III/V/B

Date: 25.9.2024

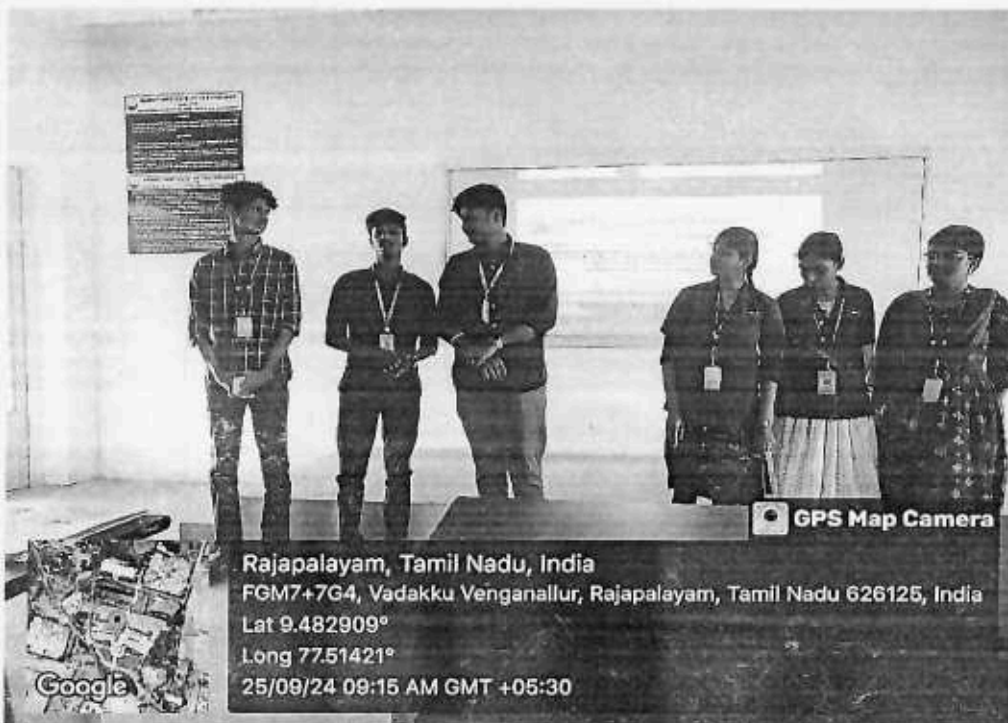
Topic : Componenets of Data and Information Security

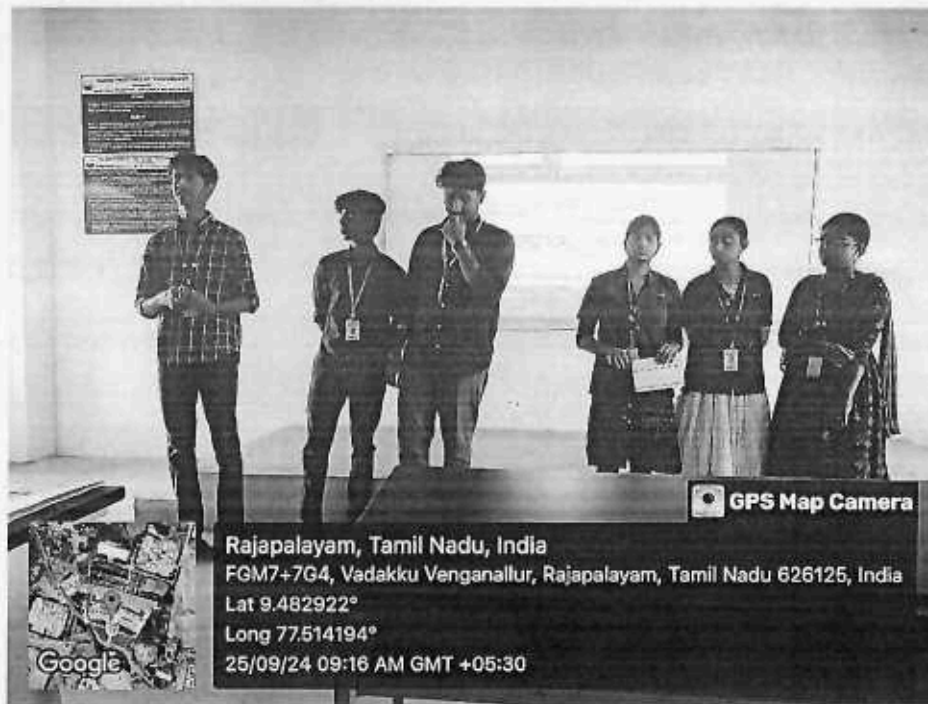
CO Coherency: CO1

Duration : 45 minutes

Role play is an interactive teaching method where students act out roles in a specific scenario or context to enhance learning. It encourages active participation and helps students understand concepts by experiencing them in a practical way.

Introduction : Course The instructor instructs the necessity of another topic and the introduction of the students, who are assigned different roles to play the components of the information system.





Component	Played by
Software	T Arun Gowtham
Hardware	G Kaleeswaran
Data	S A ThaarunKumar
People	V Prathiksha
Procedure	M Priyadarsini
Networks	K Mahalakshmi

Benefits:

- Students can enhance empathy and understanding of different perspectives.
- They can develop communication and teamwork skills.
- Makes learning more memorable and engaging.

Student Response:

Engaged Students: Students who played the role, effectively deliver their role and its responsibility.

Assisted Learners: Slow learners benefitted from one-on-one explanations provided by the students who actively participated in the role. Other students also want to participate in role play model in forthcoming topics.

Overall Enjoyment: Most students enjoyed the role play model experience, finding it helpful for understanding the basic concepts of data and information security.

Feed Back Questions

Did you find the role play engaging?

☐ Yes

☐ No

Did Your Classmates effectively communicate their role?

☐ Yes

☐ No

Did you feel the role play demonstrated the intended skills or concepts?

☐ Yes

☐ No

Do you want to participate in forthcoming innovative learning practices?

☐ Yes

☐ No

Is there any difficulty level of understanding the concept?

☐ Yes

☐ No

Pulraj L.
Faculty Incharge

mark

HoD



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

NAAC Accredited with 'A+' Grade & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Artificial Intelligence and Data Science

Academic Year 2024 – 2025 (Odd)

Degree, Semester & Branch: BTech/V/AD-B

Course Code & Title: CW3551- Data and Information Security

Name of the Faculty member (s): R M Rajeshwari

Innovative Practice Description

- **Unit / Topic:** Unit I-Components of Data and Information Security
- **Course Outcome:** CO1
- **Topic Learning Outcome:** TLO4
- **Activity Chosen:** ROLE PLAY METHOD
- **Justification:**
 - **Active Engagement:** Roleplay promotes active learning, as participants are directly involved in scenarios that mimic real-life security challenges. This active participation helps reinforce theoretical concepts by putting them into context and making the learning process more engaging.
 - **Improved Problem-Solving Skills:** Through roleplay, individuals encounter simulated security situations that require quick thinking, decision-making, and problem-solving. This helps learners develop critical thinking skills and prepares them for real-world security challenges.

Time Allotted for the Activity: 45 Minutes

- **Details of the Implementation:** It begins with setting clear learning objectives to guide the activity, ensuring participants understand the skills or knowledge they should gain. Realistic scenarios are then created based on the topic being studied, such as handling a data breach in the context of information security. Participants are assigned roles—such as a Chief Information Security Officer (CISO), hacker, or IT staff—and are briefed on their responsibilities, goals, and constraints. As the roleplay unfolds, participants make decisions, collaborate, and solve problems, applying their knowledge in a dynamic setting. Throughout the activity, the facilitator guides the group, introducing new challenges or twists and encouraging interaction. Afterward, a debriefing session allows participants to reflect on their actions, discuss outcomes, and receive feedback from both peers and the facilitator. This reflection helps learners identify strengths and areas for improvement. Roleplay also provides an opportunity for iterative learning, as scenarios can be repeated with adjustments to deepen understanding and refine skills.

• CO – PO / PSO mapping:

CO	PO 1	PO2	PO4	PO9	PO10
CO4	3	3	3	3	1

(1 – Low

2 – Moderate

3 – High)

• **PO / PSO mapped:**

Innovative practice	PSO1	PSO2	PSO3	-	-
	2	2	2	-	-
Justification for correlation	PSO1:Students will apply their knowledge of software, hardware, data, people, procedures, and networks to design and implement information systems that optimize business and engineering operations.They will apply encryption, firewalls, and database management systems to ensure secure data storage and processing. PSO2:Students will leverage cybersecurity tools, to design secure and scalable solutions, ensuring data encryption and protection from unauthorized access. They will focus on securing the physical infrastructure, including data centers and servers, to prevent breaches and sabotage. PSO3:Students will apply security software techniques such as model encryption, data storage security, and privacy-preserving machine learning to protect sensitive data and AI models .Students will also focus on educating AI practitioners about security implications, ethical concerns, and maintaining secure algorithms. Furthermore, they will design secure networks with encryption and access control to protect AI systems and data during transmission and storage.				

- **Images / Screenshot of the practice: Report Attached**
- **Reflective Critique:**
- **Feedback of practice from students and other stakeholders: Separately Attached**

❖ **Benefit of the practice:** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice) **Report Attached**

❖ **Challenges faced in implementation:**

- Implementing the role-play model while teaching components of data and information security presents several challenges. First, the complexity of concepts like data encryption, access control, and hardware security can make it difficult to create realistic and understandable role-play scenarios. Additionally, setting up these exercises requires significant resources, such as software, hardware, and realistic data sets, which can be difficult to provide in a classroom. Keeping students engaged with compelling and realistic scenarios, especially for complex topics, can also be a challenge. Furthermore, while role-playing simulates real-world situations, it may not fully replicate the pressure and complexities of actual security incidents. The varying skill levels of students add another challenge, as scenarios must be tailored to accommodate both beginners and advanced learners.